Personalized Stock Recommendation System Based on User Risk Profiling and Portfolio Optimization

A Group Project Report for the
Course Recommendation Systems (DS2803)

Submitted by:
1. Piyush Goyal (22106019)
2. Sanidhya Shyam Sagar (22106024)
3. Parth (22106026)

Submitted to:
Dr. Shilpa Verma
Assistant Professor

Abstract

- This project **aims to develop a personalized stock recommendation system that suggests investment portfolios based on an individual user's risk profile, financial goals, and investment horizon.** Unlike traditional systems that recommend generic "top-performing stocks," this system incorporates user-specific parameters to generate tailored investment strategies.
- The **system utilizes financial data such as historical returns, volatility, and market indicators to classify stocks into risk categories.** A scoring-based recommendation engine combined with portfolio optimization techniques is used to allocate assets efficiently. Additionally, the system integrates machine learning techniques for clustering and improving recommendation quality.
- The final product will be a web-based application that allows users to input their preferences and receive optimized portfolio suggestions along with performance insights and visualizations.

Keywords

Personalized Recommendation System, Stock Market Prediction, Risk Profiling, Portfolio Optimization, Machine Learning, FinTech, Recommender Systems

Introduction

- In recent years, the rapid growth of **financial technology platforms** such as **Zerodha** and **Groww** has made stock market participation more accessible to individual investors. However, most beginner investors lack the expertise required to make informed investment decisions.
- Traditional stock recommendation systems often provide **generalized suggestions** without considering the individual user's financial situation, risk appetite, or long-term goals. **This leads to suboptimal investment decisions and increased financial risk.**
- To address this issue, this project proposes a **personalized stock recommendation system** that leverages **user profiling, financial data analysis, and portfolio optimization techniques**. The system aims to bridge the gap between financial knowledge and user-specific investment planning by providing intelligent, data-driven recommendations.

Objectives

1. To design a system that analyzes user inputs (age, income, risk tolerance, investment horizon) and generates a quantitative risk score.
2. To develop a recommendation engine that matches user profiles with suitable stocks based on risk and return characteristics.
3. To implement portfolio optimization techniques to provide optimal asset allocation rather than just stock suggestions.
4. To build an interactive web-based interface that visualizes recommendations and enhances user experience.

Research/Market Gaps

We have reviewed some research papers in the domain of our project we have found some gaps in these studies which are as follow:

1. **Lack of Personalization**
   Most existing systems recommend stocks based on popularity or trends rather than individual user profiles.
2. **Absence of Portfolio Optimization**
   Many beginner-level systems suggest stocks but fail to provide optimal allocation strategies.
3. **Limited Use of Machine Learning**
   Basic systems rely on rule-based logic instead of adaptive models that improve over time.
4. **No Integration of Multiple Factors**
   Existing approaches often ignore combining financial metrics, user behavior, and market sentiment.

Proposed Solution

The proposed system aims to develop an intelligent and personalized stock recommendation framework that integrates user profiling, financial data analysis, machine learning techniques, and portfolio optimization.

Unlike traditional recommendation systems that rely on generic stock rankings, this system focuses on user-centric investment strategies, ensuring that recommendations align with individual risk tolerance, financial goals, and investment horizon.

The proposed system will follow a multi-stage pipeline:

Progress to date

- Conducted an **extensive literature review** on personalized recommendation systems and stock prediction techniques to understand existing approaches and identify research gaps.
- **Explored various financial datasets** (such as historical stock prices and market indicators) and analyzed their structure, features, and suitability for building the recommendation system.
- **Designed the overall system architecture**, defining key components including user profiling, risk assessment module, stock classification module, and portfolio optimization pipeline.
- **Developed a preliminary framework** for user risk profiling by identifying key parameters such as age, income level, investment horizon, and risk tolerance, and mapping them to a quantitative risk score.
- **Studied and shortlisted similarity and scoring techniques** (such as cosine similarity and weighted scoring models) for matching user profiles with suitable stocks.
- Performed **initial data preprocessing** steps including data cleaning, normalization, and feature selection for financial datasets.
- **Explored basic clustering techniques** (e.g., K-Means) to categorize stocks into different risk groups based on volatility and returns.
- **Created initial design prototypes** (wireframes) for the web-based interface to define user interaction flow and system usability.
- **Set up the development environment** and selected the technology stack for implementation (Python for backend, relevant ML libraries, and a web framework for frontend integration).